



PRISM ERP



Simulators Compendium

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AI Based Drill Companion (ADC)

Technical category	AI
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	0

Project description

The AI based Drill companion software has been developed to automatically identify drill posture of an indl through AI algorithm. The application may be used for drill practice, competition and drill test in various trg ests. The project has been validated by team from Arty Centre, Hyderabad. The project software application in its present state is being utilised at Arty Centre, Hyderabad. Imp features of the application are as under:-

1. Automatically identify drill posture of an indl through AI Algorithm.
2. Drill practice.
3. Drill Competition.
4. Drill test.
5. Validation and testing carried out at Arty Centre Hyderabad.



AI Based Legal Advice Sys Ver 1.0

Technical category	AI
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	0

Project description

1. AI based Software Appl for stand-alone cmptr facilitating legal referencing to sp the current manual ref practices.
2. Run on cmptrs with 8GB RAM with speaker & mic.
3. Stand-alone mode of Op with no reqmt of internet connectivity.
4. Int Query Answering Engine for user's query on Legal aspects.
5. Prompting possible addl ref alongwith the primary legal ref.
6. Facility of downloading, editing & printing draft Charge Sheets.
7. Enhanced accuracy, speed and comprehensiveness of legal proceedings due to the int of the sys.
8. State-of-the-art Natural Language Processing (AI) models used for integrating the speech recog, semantic understanding and cosine similarity to tolerate noisy inputs of user queries to be able to produce the correct legal references.



AI Based Video Svl Sys with Auto Alerts

Technical category	AI
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	0

Project description

1. AI based Software Appl can run in a stand-alone cmptr to incr efficiency of continuous real time monitoring of the AoR at Bn Svl Centre.
2. Cmptr with Graphics Card of 4 GB & 16 GB RAM.
3. Simultaneous 08 x live video feeds can be processed in near Real Time & the results displayed on the screen.
4. Integration of Day & Ni Svl devices. Live video feeds from LORROS, HHTI, CCTV camera and PTZ camera can be processed simultaneously.
5. Stand-alone mode of Op without reqmt of internet.
6. 11 different tgt cl viz Arty Guns, Heptrs, Drones, Tks, Pers, Rif, Animals, Trucks, Mor, Cars, Two wheelers to raise the alarm.
7. Unique coloured bounding boxes specific to each tgt class as well as nomenclature of the tgt class (example: Heptrs, Drones, Tks etc) are displayed on the screen. An audio alarm is also raised when any one or more of the video feeds detects any one or more types of tgt classes.
8. State-of-the-art Cmptr Vision (AI) models used for image processing and near real time tgt detection.



VR based Mil Drone Flying Sml (VRMDFS)

Technical category	AR/ VR
Year of completion	2025
Arm/Service	Not recorded
Proliferation cost (lakhs)	7.5

Project description

1. (VRMDFS) is a sophisticated trg sml based on Virtual Reality, designed to mimic the experience of operating unmanned aerial vehicles (UAVs) viz. Quadcopter, Hexacopter, Fixed Wing platforms for tps of IA to gain first time near real experience in operating these ae platforms without damaging the real eqpt.
2. The sml provides realistic simulations of drone ops, including takeoff, flight, obstacle negotiation, landing, hovering as well as various other msn scenarios in eight types of trns to include HAA, mtns, plains, semi desert, desert, costal, BUAs, Jungle trns.
3. A 1:5 config for trg five trainees concurrently in multi trg model and trg an individual trainee in solo model is avlb in the sml.
4. The sw for the sml has already been devp and is being proliferated to the units of IA.

Software is provided free of cost. The cost is only for the COTS HW.



VR based Multi Mode Dvg Sml (VRMMDS)

Technical category	AR/ VR
Year of completion	2024
Arm/Service	Not recorded
Proliferation cost (lakhs)	7.5

Project description

Various features of the sml sw are as follows: -

1. Enables anytime dvg trg.
2. Instr can customize and apply various scenarios and contingencies in real-time.
3. Eight vehicle variants i.e.
 - Maruti Gypsy
 - Maruti Ciaz
 - Maruti Dzire
 - Tata Safari Storm
 - Tata 2.5 Ton
 - HMT 6x6
 - HMT 8x8
 - ALS 8x8 Super Stallion
4. Training can be provided in Seven different trns and different wx conditions in both day and night conditions.
5. Saving of fuel and resources employed in trg.
6. Provides After Action Report (AAR).



Virtual Reality Based 3D Trn Visualisation (VR-Trn)

Technical category	AR/ VR
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	3.5

Project description

The sys will facilitate user in area visualisation using 3-Dimensional Geo-Specific trn in VR envt reducing the reqmt of physically going on the grnd. The salient features are as under:-

1. Geo Specific Trns of complete India and min 30 Kms in depth with respect to Indian borders with neighbouring countries.
2. Totally Offline; NO reqmt of internet.
3. Upload & Overlay maps of areas of interest.
4. Display Latitude, Longitude, Mil GR and Alt.
5. Placing 3D tac obj from a predefined library.
6. Placing svl device to show view shed.

Software is provided free of cost. The cost is only for the COTS HW.



Virtual Reality Based 84mm RL Firing (VR-RL) Sml

Technical category	AR/ VR
Year of completion	2021
Arm/Service	Inf
Proliferation cost (lakhs)	3.5

Project description

To train 84mm RL firer on firing aspects on tac tgts in realistic immersive scenarios. The salient features of the trg sml are as under:-

1. Annual Rg Course trg module to conduct trg on firing of RL as per annual rg course in fwg trns:-
2. Desert.
3. Semi Desert.
4. Mountainous.
5. Scenario Based Trg on firing of 84 mm RL on static & mob tgts in realistic tac scenarios.
6. Strap on mod on 'in service RL' for higher sense of realism.



Virtual Reality Based Ac Recog (VR-ACR) Trg Sml

Technical category	AR/ VR
Year of completion	2021
Arm/Service	AAD
Proliferation cost (lakhs)	12

Project description

The sml facilitates simultaneous trg of 20 x AAD observers on ac comparison and recog techniques in a virtual envt. The Sw comprises of various functionalities/modules as listed below:-

1. Instr console to create ex, monitor trg and eval trainees.
2. Six trns to incl Plain, Desert, Semi-Desert, Mountains, Jungle & Costal Region
3. Facility to train/ eval trainee on recog of 55 different acs
4. Facility to create/ edit multiple ac flt path.
5. Option to select and att ac to the path singly or in a gp.
6. Option to dply single or gp of obs along the route.
7. Flt parameters like speed & alt can be set by instr.
8. Option to set various envt conditions like time of day, rains, fog, wind, storms, etc.



Virtual Reality Based Ae Del (VR-AD) Trg Sml (AN 32 and MI 17)

Technical category	AR/ VR
Year of completion	2021
Arm/Service	Inf/ ASC
Proliferation cost (lakhs)	5

Project description

To facilitate Offrs, JCOs and OR of the IA in imparting trg on Ae Del to incl parachute packing, lashing of loads on platform and dropping from different ac (fixed wg & rotary wg i.e. AN 32 and MI 17) under varied trn conditions using VR tech. The Sw comprises of functionalities as listed below:-

1. Components/ Eqpt Iden.
2. Pre Packing Prep and Insp Procedure.
3. Packing & Lashing of Load.
4. Insp to be done before final drop.
5. Emplaning and Deplaning drills.
6. Drills involved in drop of loads.
7. Incr difficulty level by changing envt & wx conditions.



Virtual Reality Based Obsn Post End Trg Sml (VR-OP) with TIIOE Mockup

Technical category	AR/ VR
Year of completion	2021
Arm/Service	Arty
Proliferation cost (lakhs)	16

Project description

Devp with an aim to train the complete Arty OP Party in Obsn, Acquisition & Engagement of tgts, trg on handling of TIIOE & OP Menu of Proj Shakti ETC using VR in an immersive envt. The salient features are as under:-

1. Trg on 14 types of shoots/ procedures through 11 types of arty eqpt with HE, smoke & ill amn.
2. Trg on Conventional shoots, Shoots with TIIOE & TIIOE integrated with ETC.
3. TIIOE Mockup incl ARTIMLR, LRF, Goniometer and other accessories to impart trg on handling of TIIOE.
4. Practice on Six Geo Specific trns akin to arty fd firing rgs having variable difficulty to incl large apex angle shooting, engagement of tgt on crest, moving tgt, etc.
5. Variable envt factors to incl Rain, Fog, Storm etc and day/dusk/ni shooting.



AR Based Adv Mannequin Sys (AR-AMS)

Technical category	AR/ VR
Year of completion	2020
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	60

Project description

1. The sml facilitates trg of MO, Paramedics, BFNAs and all rks of the IA on paramedics and Combat Trauma Life Sp (CTLS) procedures.
2. The sml consists of an instrumented Adv Mannequin Sys (AMS) which is akin to human body along with associated accessories and COTS hardware.
3. In addn, AR headset incr the realism in the trg by overlaying injuries onto the AMS.
4. The paramedics trg module facilitates in trg of trainee on basic paramedic procedures.
5. In CTLS module, the trainee is aligned to a scenario for which animated video of the scenario is played. Subsequently the Aim/ learning obj is explained followed by assessments to be done by the trainee is explained.
6. In the end, the trainee is guided by animated video, descriptions on the screen and headset and audio cues to perform procedures on the patient. The trg module help to eval the trainee on paramedics and CTLS procedures.



Virtual Reality Based Comb Med Care (VR-CMC) Sml

Technical category	AR/ VR
Year of completion	2020
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	3.8

Project description

1. The VR CMC has been devp to train MO and Paramedics on Human Anatomy in addn, BFNAs and all rks of the IA on Comb Med Care procedures in an immersive envt.
2. The familiarisation module consists of fwg sub modules:-
 - Human Anatomy and Physiology
 - Patient Safety Care
 - Critical Care Transport
 - CBRN
3. The CMC module can be used for trg of trainees on imparting first aid in case of various med emergencies.
4. 21 different scenarios have been modelled in three sub trn configurations viz Desert, Jungle and Snow wherein various scenarios are build up where the model sustains injury and trainee is trained to give first aid accordingly.
5. The eval module facilitates to eval the trainee on steps to be followed while imparting first aid and is accordingly eval.

Software is provided free of cost. The cost is only for the COTS HW.



VR Based Engine Familiarization for Tk T-90 Sml (VR-T90)

Technical category	AR/ VR
Year of completion	2020
Arm/Service	Armd/ AOC/ EME
Proliferation cost (lakhs)	3.5

Project description

To train A veh tech, crew and SKT on engine familiarisation and maint of the engine sub assys in an immersive envt. The software comprises of various functionalities/ modules as listed below:-

1. Engine Familiarisation.
2. Parts Familiarisation with CAT Part No.
3. Familiarisation to Cooling and Lubrication Sys.
4. Air Distribution Timing Setting Procedure.
5. Valve Timing Setting Procedure.
6. FIP Timing Setting Procedure.

Software is provided free of cost. The cost is only for the COTS HW.



Virtual Reality Based Intermediate Gear Box of Tk (VR-IGB)

Technical category	AR/ VR
Year of completion	2018
Arm/Service	EME
Proliferation cost (lakhs)	3

Project description

The VR based visualiser provides a comprehensive realistic trg on IGB in a virtual, immersive envt to address the trg need of the tk crew and techs. The salient features of the sml are as under:-

1. Explode/Implode Mode facilitates trainees to see the IGB open out (dis-assemble) or close (assemble) in a sequential manner.
2. Necessary SMT's will appear at places wherever reqd to open/close any part.
3. Transparent Mode provides functioning of various gears of IGB in transparent mode.
4. Op Mode shows a transparent model of Tk T-90 demo various drives of IGB and its integration with other components like Engine, Starter Generator, Fan, SGB etc.
5. Eval Mode facilitates testing of trainee on assembling and disassembling procedures.

Software is provided free of cost. The cost is only for the COTS HW.



Simple Driving Sml (SDS)

Technical category	Driving Smls
Year of completion	2014
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	13

Project description

1. Promulgates the penetration of sml to field units of IA by providing free of cost Desktop Based Driving Simulator solution. This setup works with COTS Gaming Hardware (Logitech G-29).
2. In house software devp by SDD. Ex with varying levels of difficulty and envt effects across various trns.
3. Playback facility provided in the sw.
4. Realistic graphics with immersive features implemented in the ex.
5. 20 x prefed exercises with scope to customize new ex setups.
6. Simple to use, maintain and op COTS gaming controller used.
7. All fn of the veh instrument panel and switches tapped to align drivers to its use.
8. Set up can be run using a simple table and chair or ergonomically designed veh frame.



Tata 2.5 Ton Driving Sml

Technical category	Driving Smls
Year of completion	2014
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	Not recorded

Project description

1. Provides realistic trg on TATA 2.5 Ton for raw trained drivers in familiarizing with controls of the TATA 2.5 Ton vehicle under various driving conditions (terrain, visibility, weather, load, traffic etc) and assessing their performance.
2. Interactive Instructor Stn have facility to connect Multi 'B' Veh driving smls.
3. Presents real feel like interactive visuals with interactive veh sounds to the dvr.
4. Driving under varied vis conditions.
5. Fabricated mockup with 3 DOF motion platform made up of fiber reinforced plastic (FRP) material with original dash board controls, brake, accelerator, clutch, gearshift lever and steering.
6. Visuals projected on screen through projector for Front and Rear view.
7. Cvy Driving.
8. Instr has fwg facility: -
 - Start, stop and restart a session.
 - Varying diurnal and weather effect
 - Injection of faults
 - Performance Eval
 - Report generation.
 - On-Line help
 - Realistic math model of veh dynamics.



Universal Driving Sml (UDS)

Technical category	Driving Smls
Year of completion	2014
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	24

Project description

1. Instr Stn/ Display unit has facility to connect Maruti Gypsy, TATA 2.5 Ton and Ashok Leyland Stallion dvr stn.
2. he sml presents real life like interactive visuals with interactive veh sounds to the dvr.
3. Driving under varied vis conditions.
4. It is a fabricated mockup with a soft panel dash board, original op cont, brake, accelerator, clutch, gearshift lever and steering mechanism.
5. Visuals are projected on three 32" LCD TV screens and a 15" screen through a four port AMD Radeon 7800 Series graphic card.
6. The sml is based on Roll On Roll Off concept.



T-90 Driving Sml

Technical category	Driving Smls
Year of completion	2013
Arm/Service	Armd
Proliferation cost (lakhs)	90

Project description

1. User friendly Interactive Instr Stn.
2. Facility to create ex.
3. Running fault injection for realistic trg.
4. Continuous monitoring of driver's actions.
5. Realistic audio effects.
6. Test s/w for fault diagnosis.



BMP-II Driving Sml

Technical category	Driving Smls
Year of completion	2008
Arm/Service	Mech Inf
Proliferation cost (lakhs)	Not recorded

Project description

1. Missile Tracer Flare (MTF) unit to simulate Konkurs missile in flight.
2. Firing post with connected optics and facility to simulate fire of Konkurs ATGM, 30mm Cannon, 7,62 PKT and 81mm Smoke Grande Discharger (SGD).
3. Gunnery compartment housing all controls and sights as in the turret station of a BMP-2.
4. Computer with necessary software for a high level of data generation to simulate various facilities.
5. Digitised Projection System to project through the sights the terrain / target view duly integrated with battlefield effects.
6. Realistic and functional replication of ATGM System, 30mm Cannon, 7.62mm PKT and 81mm SGD.
7. Interface between the Computer, Instructor Console and Firing Post.



OSA AK Fg Sml

Technical category	Firing Smls
Year of completion	2025
Arm/Service	Not recorded
Proliferation cost (lakhs)	24

Project description

1. It is a trg tool to replicate the wpn sys's functionality and op for pers trg.
2. This sml aims to provide realistic scenarios for oprs to learn how to detect, tr, and engage tgts, incl aircraft, heptrs, and drones, without the need for live msls or real-world conditions.
3. This sml will be used to train the opr on the functioning and op of Assistant Crew Commander (ACC), Search Operator (SO) and Range Operator (RO) panels.
4. This sml adequately trains oprs on **Mode I: Automated Target Tracking Mode, Mode II: Target Designation Mode & Mode III: Aided Tracking Mode** of OSA-AK wpn sys while simulating various deg of difficulties, dynamically introducing multiple / single tgt allowing to vary the flying speeds.
5. It also caters for slow moving tgts and low flying tgts. This sml op in Trg mode, Normal Mode & Eval mode.
6. It also allows the instr to train the trainee on carrying out various ECCM drills reqd to counter ECM carried out by en aircrafts, thus providing realistic trg experience.



Untethered Recoil Kits for Integration with Firing Sml

Technical category	Firing Smls
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	5

Project description

The simulator facilitates in conduct of basic squad post training of soldiers and simulates all range course exercises for firing upto a range of 400m, with untethered recoil kits for simulating recoil of wpn being fired. Few imp features are as under:-

(a) Enables devp of simulators for in-service and new genr wpns.

(b) Untethered Recoil kits for fwg Seven (07) wpns: -

- 9mm Browning Pistol
- 9mm Carbine Sub Machine Gun
- 5.56mm INSAS Rifle
- 5.56mm INSAS Light Machine Gun
- 7.62mm AK-47 Rifle
- 7.62mm Sig Sauer 716 Rifle
- 7.62mm Dragunov Sniper Rifle

(c) Pneumatic Op with inert gases

(d) Supports integration with existing Drona & IWTS simulators.

Production Cost for 1 wpn kit (recoil kits, laser module, 02 x magazine) - ₹5 lakhs



0.22 Rif Firing Sml

Technical category	Firing Smls
Year of completion	2021
Arm/Service	NCC
Proliferation cost (lakhs)	Not recorded

Project description

1. Modular Design with four firing lanes.
2. Use of eye safe laser to shoot on virtual targets rendered on screen as per actual distance.
3. Facility to Zero wpns.
4. Realistic evaluation of firer.
5. Firing in lying, kneeling and standing.
6. Five target sizes.



AGLSDS (ATGM Grnd LR Skill Devp Sml)

Technical category	Firing Smls
Year of completion	2018
Arm/Service	Inf
Proliferation cost (lakhs)	17.1

Project description

To provide holistic trg to the missile crew on Konkurs based ATGM Lr.

1. Dual Mode Op. It can impart skilled based trg as well as expose the crew to wk in an integrated tac envt.
2. Thermal Modelling. Thermal Modelling has been specifically devp with different shades to achieve near realistic view of Thermal Imaging (TI) Sight.
3. Varied Trns & Scenarios. The sys provides flexibility to choose from various types of trns (desert, plains, High Altitude Area etc) and allows creating scenarios of varied complexity as per the skill levels of the trainees.
4. Very High Level of Realism. The simulator has elevation and bearing control, sight view, obturation effect and Missile path depicted very near to the real effects in the actual sys.



Multi Utility Unmanned Ground Vehicle (MU-UGV)

Technical category	Robotics
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	25

Project description

The multi utility Unmanned Ground Vehicle (MU-UGV) has been developed in house by SDD. The sys can carry a ld of 100 kg, can be utilised for cas evac from battle fd and to remove the Unexploded Ordnance (UXO) through a Robotic Arm. Few imp features MU-UGV are as under:-

1. Ld caring cap upto 100 kg.
2. Cas evac.
3. Disposal of unexploded Shell from the firing rg with the help of Robotic Arm.
4. Endurance of four hours in continuous op in Bty mode.
5. Video & Control signal txn rg of 500m LOS. Can be enhanced upto 3000m LOS, as per op reqmt.
6. Gradient negotiating capb 25 Deg. Can be enhanced upto 60o degrees as per op reqmt.
7. Max Speed - 08 Km/h.
8. Real time video feed from front and rear Camera 3MP resolution 1080p.



Tactical Combat Robot (TCR)

Technical category	Robotics
Year of completion	2022
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	94

Project description

1. The Tactical Combat Robot (TCR) is an AI-enabled robotic platform capable of performing surveillance, reconnaissance and limited engagement in high-risk combat zones, thereby minimizing soldier exposure.
2. The system integrates advanced technologies such as Artificial Intelligence, Robotics and Computer Vision to enhance operational efficiency in Counter-Insurgency/ Counter-Terrorism operations, border surveillance and conventional warfare.
3. The TCR features a rugged tracked platform with a 60 kg payload, 8 km/h speed, and 3-hour endurance. It carries a 5.56 mm INSAS LMG, day/night camera with 30X optical zoom, thermal vision, laser range finder, GPS, and IMU for real-time situational awareness.
4. Communication between the robot and Ground Control Station (GCS) is through RF modules offering up to 3 km range and 100 Mbps throughput.
5. A customized GUI-based control software displays live video feed, GPS location, target tracking, platform health and weapon status, enabling precise control and coordination.



Cockpit Sml for Cheetal Helicopter

Technical category	Misc
Year of completion	2024
Arm/Service	Not recorded
Proliferation cost (lakhs)	Not recorded

Project description

1. To train Pilot & Co-Pilot on drills like pre-flight check-up, engine starting & stopping procedure, rotor engagement and disengagement, and vent start drill and emergency procedures on near realistic models of Cheetah and Chetak helicopters.
2. Run-Time faults specific to a particular drill during the pre-flight checks can be induced by the instructor.
3. In addition, the trainee's performance report, based on the weighted-average model, is presented to the instructor to facilitate evaluation.



ARV WZT-3 Rec Trg Sml

Technical category	Misc
Year of completion	2022
Arm/Service	Armd & EME
Proliferation cost (lakhs)	25

Project description

1. To Provide holistic trg to the crew on drills & procedures to op ARV WZT-3 to refine the skills of crew in carrying out recovery ops in varied terrain & tactical conditions.
2. Mockup with controls of actual Dvr compartment similar to original eqpt.
3. Provision to carryout realistic rec OP for different types of cas in varied terrain and envt conditions.
4. Interactive instr stn with all the cont for making trg challenging incl insp of run time faults.
5. Interactive 3D models of A Vehs & rec resources.
6. 08 x Winching drills incl under water rec with modules for rec, earth moving & maint.
7. Quantified eval report generated on culmination of each trg session helps in gauging the trg imperatives gained by the trainee .
8. Realistic sound cues for op of the eqpt incorporated in the Sml for more realistic trg.



BLT T-72 Br Trg Sml

Technical category	Misc
Year of completion	2021
Arm/Service	Engr
Proliferation cost (lakhs)	23

Project description

The BLT T-72 is a potent Aslt Br launching eqpt emp by specialist Engr Regt for sp Mech Coln rush drills. The sml is a potent sml trg soln enabling the trainee to learn and update op skills on this newly inducted eqpt. The trn is designed to expose the trainee on op emp of the eqpt. Interactive Instr Stn enable the instr to create/design & run the ex as per the desired trg std reqd to be achieved for the trainee. On culmination of each ex a detailed eval report is being generated. The sml consists of the fwg imp feature: -

1. One indicative mock up units for dvr cabin with instrumented cont for launching/ de-launching op.
2. Portable joystick units to op the br, as part of the original eqpt.
3. Crew drill to be depicted for better assimilation of op proc involved in use of actual br ops.
4. Varied climatic conditions.
5. Trainee performance module.



Radio Telephony Sml

Technical category	Misc
Year of completion	2021
Arm/Service	All Arms/ Services
Proliferation cost (lakhs)	3.5

Project description

To provide realistic trg to the opr of RS STARS V on RT procedures and basics of RS without violating the EEP policy.

1. Capb to connect upto 15 Oprs in a net.
2. Perform all basic Radio Telephony functions (e.g feeding of freq, Selective call, Secure call, Hailing etc).
3. Instr has the facility to intro different types of Jx (spot and barrage jx).
4. Capb to put one or more opr console out of comn simultaneously.
5. Audio record and monitor console activities carried out by any selected opr.
6. Provn to make audio comn and send msg (upto 300 words) individually/ selected/ broadcast mode to all the trainees.
7. Maint record of trg where in pers particulars, duration, dt & time, type of ex and activity carried out by the trainee is recorded.
8. Opr Stn can communicate upto a dist of 30 ft from the Instr Stn, wirelessly using a router.



HRV AV-15 Trg Sml

Technical category	Misc
Year of completion	2019
Arm/Service	Armd & EME
Proliferation cost (lakhs)	26.06

Project description

1. Provides realistic trg on HRV AV-15 to raw/ trained dvrs & to familiarize them with rec cont under varied conditions (trn, visibility, weather, load, etc.) and assess their performance.
2. Provides hands on trg to rec dvrs and oprs on critical aspects of any rec op in terms of assmt of eqpt cas, trn & grnd conditions.
3. Mock-up with cont which are very similar to original eqpt.
4. Provision to carryout realistic rec op for different types of casualties in varied trn and envt conditions.
5. Interactive Instr Stn with cont for enhanced trg.
6. Trainee evaluation module for After Action Review
7. Two separate mock up units for dvr & rec cabin with actual instrumented controls.
8. Rec ex for majorly all veh/eqpt held in the inventory of IA.
9. Crew drill are depicted for better assimilation of op proc involved in use of actual rec ops.
10. Varied climatic conditions.
11. Trainee performance module.



Sarvatra Br Sml

Technical category	Misc
Year of completion	2019
Arm/Service	Engr
Proliferation cost (lakhs)	27.82

Project description

1. Sml is an effective trg aid for trg of crew of all the Aslt Engr Regt, Armd Div Engr Regt, RAPID Engr Regt, Indep Engr Sqn and associated Cat A/B trg est for the Corps of Engrs
2. **Trainer for Complete Crew.** The sml enables skill level enhancement for dvr by providing an already matured Universal Driving Sml (UDS) adapted for the prime mover of this eqpt. It also provides trimmed mock up unit for the rear dvr cabin along with launching and delaunching mock up units for real feel of op of this aslt br. The crew of four pers incl cdr, op & dvr can be trained effectively on this sml.
3. **Varied Trn & Scenarios.** Sys provides flexibility to choose from 36 various trns. It has 10 different profiles each for rivers, canals, DCBs and allows creating scenarios of varied complexity as per the desired skill levels expected from the trainee.
4. **Realistic 3D Modelling.** Realistic 3D modelling of the prime mover and the br sys as been depicted in the Comptr Generated Imagery (CGI) of the sys incl rendering of animated crew drills for better assimilation by the trainees.
5. **Ex Set Up.** The ex have been created as per the original time involved in exec of various op as per tac emp of the original eqpt.
6. **Performance Eval.** Detailed After Action Report (AAR) is gen by the sys indicating the exec of all the seq of op involved in operating the eqpt.



Desktop Based Rec Sml (DBRS)

Technical category	Misc
Year of completion	2018
Arm/Service	EME
Proliferation cost (lakhs)	0

Project description

1. Provides realistic trg to the dvr and rec opr on ARV WZT-3, HRV AV-15 and LRV in various trns and under various levels of difficulties.
2. In house software devp by SDD
3. Ex with varying levels of difficulty and envt effects across various trns.
4. Module for comprehensive familiarization with all aspects of rec trg.
5. All fn of the veh instrument panel and switches tapped to align drivers to its use
6. Multi-Configuration Scenario for customized ex creation
7. No Addl infra Reqmt
8. Pre trg before actual rec on grnd.



ALS Based Light Recovery Vehicle (ALS LRV) Sml

Technical category	Misc
Year of completion	2016
Arm/Service	EME
Proliferation cost (lakhs)	10

Project description

1. Designed for rec of mobilised/blogged down/ ditched 'B' veh or evacuation of damaged veh and also to provide eff trg to ALS based LRV opr.
2. Complete system consists of look alike, full scale ALS LRV operator console mock-up, an Interactive Instructor Station (IIS), stn for Computer Generated Imagery (CGI), realistic sound module and a projection sys to project the scenario to the trainee.
3. Scenario can be prepared by instr for each trainee and envt variables can be varied while on run any time by the instr.
4. Availability of multiple 3D synthetic trns.
5. Availability of all type of B veh cas for practice.
6. Facility of familiarization with consoles as well as starting, operating, rec and stopping drills.
7. Facility to create and store new ex by instr.
8. More than 250 permutations of exs.
9. Based on exact drills and type of rec ops.
10. Facility of load calculation by mathematical means and accordingly resource selection.
11. Realistic sound effects.



Tk Tp Tac Trg Sml (T4S)

Technical category	Misc
Year of completion	2016
Arm/Service	Armd
Proliferation cost (lakhs)	Not recorded

Project description

T4S is a PC based tactical trainer which aims at practicing battle drills of tank commanders at a troop & squadron level. Each PC depicts a virtual tank with provision for driver, commander, gunner sights controlled through keyboard & joystick. These tanks are exercised against enemy elements & field defense over various tactical situations which can be created by the instructor in different types of terrains. The project has 14 tanks or 15 ICVs operating on a networked & integrated High Level Architecture (HLA) platform. Special effects of visibility, weather, firing sound, inter-communication, virtual soldiers & evaluation are available. User is required to cater for high end system while software is shared without any cost accruals. Few important features are as under :-

1. The architecture is entirely plug and play and addl entities can be seamlessly added to extend the concept to CT level and beyond.
2. Dynamic cont of own/ en forces.
3. Library of A Vehs for creation of own and en forces.
4. Time-compression and dynamic attrition cont.
5. Obj library with landmarks for creation of own features.
6. Trainee Performance and After Action Review.

Software shared without any cost accruals.

